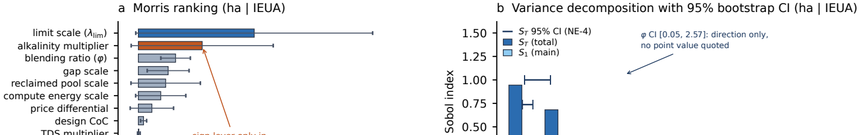

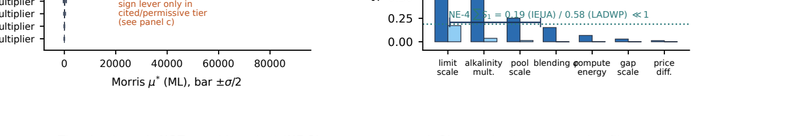

strict tier tightens TDS and silica while alkalinity stays 350
– so the binding set becomes {silica, TDS} and the lever disappears

Data: A34_WP5/wp5_morris.csv, wp5_sobol.csv, wp5_extreme_cases.csv, wp5_extreme_summary.json (panels a,b,d); A38_simulated-review/ne4_sobol_bootstrap_summary.json (b, 95% CI) and ne2_panel_robustness_summary.json (c).

Morris/Sobol panels a,b: Saltelli design on config ha|IEUA; LADWP configs report NOT_IDENTIFIABLE and are NOT plotted. NE-4 CI: 400 bootstrap replicates, N_base=128, d=7; n_eff = 117 (IEUA) / 128 (LADWP) after dropping degenerate points. NE-2: 2 complete-species samples x 3 limit tiers x 2 domains, phi=0.6. Extremes: n_eff = 64 of 64 windowed scenarios after the dual gate. ZERO CONVENTION: |dS_pot| < 1e-6 ML counted as 0.

LIMITATION: (i) Sum S1 = 0.19 (IEUA) / 0.58 (LADWP) is far below 1, so point rankings describe only part of the variance; (ii) phi's CI [0.051, 2.568] is so wide that only its direction, never a point value, may be quoted; (iii) the alkalinity sign lever is a sensitivity boundary WITHOUT observational support (IEUA 140 mg/L, LADWP 7.1 mg/L) and it does NOT survive the strict make-up tier of NE-2 - it is a lever identified within an explicitly bounded limit domain, not a universal mechanism; (iv) extreme-event windows use driver-only forcing and a single-city domain.